automatically forward cryptographic operations to the KMS via the PKCS#11 API.

> **TIP** Java's built-in PKCS#11 cryptographic provider only supports a few algorithms, many of which are old and no longer recommended. A KMS vendor may offer their own provider with support for more algorithms.

Listing 11.13 Accessing a KMS through PKCS#11

```
var keyStore = KeyStore.getInstance("PKCS11");    #A
var keyStorePassword = "changeit".toCharArray();    #A
keyStore.load(null, keyStorePassword);    #A

var signingKey = (PrivateKey) keyStore.getKey("rsa-key",
#B
        keyStorePassword);    #B

var signature = Signature.getInstance("SHA256WithRSA");    #C
signature.initSign(signingKey);    #C
signature.update("Hello!".getBytes(UTF_8));    #C
var sig = signature.sign();    #C
```

#A Load the PKCS11 keystore with the correct password
#B Retrieve a key object from the keystore
#C Use the key to sign a message

PKCS#11 and hardware security modules

PKCS#11, or Public Key Cryptography Standard 11, defines a standard API for interacting with hardware security modules (HSMs). An HSM is a hardware device dedicated to secure storage of cryptographic keys. HSMs range in size from tiny USB keys that support just a few keys, to rack-mounted network HSMs that can handle thousands of requests per second (and cost tens of thousands of dollars). Just like a KMS, the key material can't normally be accessed directly by clients and they instead send cryptographic requests to